

# RAUDEL ALEJANDRO GÓMEZ MOLINA

## Computer Scientist | Researcher

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## EDUCATION

Bachelor's Degree in Computer Science

University of Havana – Cuba

Nov 2021 – Mar 2025

- Distinction:** Summa Cum Laude and Scientific Merit Award.
- GPA: 4.84/5.00
- Core areas: Algorithms, Data Structures, Algebra, Discrete Mathematics, Databases, Distributed Systems, Programming Languages Theory, Artificial Intelligence.
- Thesis: *An Approximation to the Language of All Satisfiable Boolean Formulas.*

## TEACHING EXPERIENCE

Assistant Professor

University of Havana

Apr 2022 – Mar 2025

Havana, Cuba

- Subjects:** Data Structures and Algorithms, Algebra.
- Mentored undergraduate students; designed problem sets and ICPC-style exercises for Competitive Programming training.
- Designed and helped implement lectures and practical classes for the course programs.

## RESEARCH & ACADEMIC PROJECTS

An Approximation to the Language of All Satisfiable Boolean Formulas

Undergraduate Thesis – University of Havana

2024 – 2025

- Links SAT (NP-Complete) with formal language theory to construct the language of all satisfiable Boolean formulas.
- Two strategies: (1) transduction of a Copy language variant, proving NP-Hardness of the word problem for closed formalisms; (2) a Range Concatenation Grammar recognizing this language, showing RCGs capture all NP problems as formal languages.

## PROFESSIONAL EXPERIENCE

Full Stack Developer

jobs2live

Sep 2023 – Jan 2024, Mar 2025 – Present

Madrid, Spain

- TypeScript, React, Next.js, C#, .NET, PostgreSQL, OData, Playwright, Web Scraping.
- N-layer architecture with CQRS pattern; database modeling and LinkedIn scraping bots.

Mobile Developer

Business Seed & Bizne MX

Feb 2024 – Present

Mexico DF, Mexico

- Dart, Flutter, Getx.
- Hybrid Android/iOS apps with Clean Architecture; push notifications, maps, QR scanning.

## TECHNICAL SKILLS

C/C++PythonGoRustAlgorithmsData StructuresCompetitive ProgrammingProblem SolvingAlgorithmic ThinkingResearchLaTeXGitDockerPostgreSQLReactTypeScriptC#DartFlutter

## RESEARCH INTERESTS

- Algorithms & Computational Complexity
- Competitive Programming
- Artificial Intelligence
- Software Engineering & Architecture
- Programming Language Theory

## ACHIEVEMENTS

- Summa Cum Laude** – Bachelor's Degree in Computer Science, University of Havana (2025).
- Scientific Merit Award** – Recognized for outstanding undergraduate thesis.
- Silver Medal** – ICPC Caribbean Finals 2023.
- 12th place – ICPC Caribbean Finals 2022.
- 10th place – ICPC Caribbean Finals 2021.
- National Pre-selection in Mathematics Olympiad (2 years).

## LANGUAGES

- Spanish** – Native
- English** – Technical reading proficiency

## SELECTED PROJECTS

Travel Agency

Distributed Systems Course Project

2023

- React, Antd, .NET, MySQL, OData.
- Multi-agent distributed platform for tourists and travel agencies.

Teslo Shop

E-commerce platform

- React, Next.js, Tailwind CSS, PostgreSQL.
- Full e-commerce with authentication, payments and product management.